

CubeSat Electrical Power System with MPPT Solar Charging and I²C Telemetry

Applications

- centisat 1U–3U CubeSat platform (MSU senior capstone)
- University and educational CubeSat power systems
- General-purpose Li-ion + MPPT solar bench development platforms

Features

Topology

- Body-mounted **4S solar array** input — 4× SM141K10TF modules per face, 4 faces parallel
- **2S2P LG INR18650 MJ1** Li-ion battery pack (7.0 Ah / 50.4 Wh nameplate)
- **7.27 V nominal** battery bus (8.4 V max, 5.0 V min)
- Two independent synchronous-buck rails: **+5 V** and **+3.3 V**

Electrical

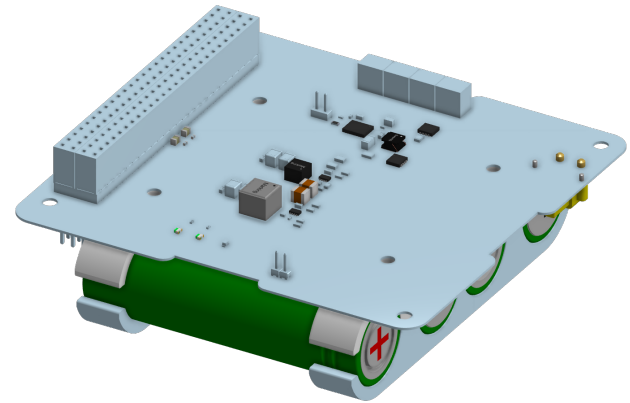
- **LTC4162-L** Multi-Cell MPPT Li-ion Step-Down Charger
 - Up to **3.2 A** charge current
 - **35 V** absolute max input
 - I²C telemetry: V_{IN} , V_{BAT} , I_{CHG} , T_{BAT} , SoC, state register
- **TPS62933F** synchronous bucks (×2, one per output rail)
 - **3.8–30 V** input range, up to **3 A** continuous
 - Per-rail EN/UVLO programmable via resistor divider
 - 200 kHz – 2.2 MHz programmable switching
- Solar Voc up to ~28 V (4S cold), Vmp ~22 V — full margin to LTC4162 35 V ceiling
- Schottky-OR'd solar inputs (per-panel STPS1L30MF blocking + per-cell-array bypass)

SWaP

- **PC/104-compatible CSKB stack-bus** form factor (~96 × 90 mm)
- Single-board EPS — no daughter cards required
- Mass: < 100 g (estimated, final TBD)
- Stack-bus integration via H1/H2 headers per centisat CSKB pinmap

Environment

- Operating temperature: **–20 °C to +60 °C** (preliminary)
- Battery charge-disable below 0 °C via LTC4162 NTC (planned)
- 1 oz copper, FR-4, 4-layer (controlled-impedance candidate)
- 6 V/m EMC margin (preliminary, TBD on full chamber sweep)



Project Information

Field	Value
Project	centisat EPS
Current revision	Rev A (prototype, on bench)
Next revision	Rev B (in design)
Repository	github.com/ngrabbs/centisat
Owner	MSU CubeSat (capstone)

BRING-UP STATUS

Rev A is in active bench bring-up. LTC4162 charge path verified; TPS62933F rail bring-up in progress (see `phase1_validation.md`). **Not flight-qualified.**

Protection & Safety

- **Remove-Before-Flight (RBF)** jumper for hard system inhibit
- LTC4162 NTC-based over/under-temperature charge inhibit
- Schottky blocking diodes prevent panel reverse current at night
- Per-cell-array bypass diodes on each panel string
- Charger-centric battery protection (independent BMS deferred)
- Soft-start on both bucks (33 nF C_{SS} , ~5 ms ramp)

Interface

- **CSKB stack-bus** (H1/H2) — locked per centisat CSKB pinmap
- **I²C** to internal housekeeping unit (IHU) — charger telemetry
- **+5 V**, **+3.3 V** rails distributed on stack-bus
- Battery bus and GND on stack-bus
- Solar panel headers (5-pin, one per face) — dedicated, off-stack